



Bachelor thesis

Homogenization in rock mechanics

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Homogenizace v mechanice hornin

Abstrakt

Tato bakalářská práce se zaměřuje na numerickou homogenizaci mechanických vlastností krystalických hornin, jejichž heterogenní struktura je popsána diskrétní puklinovou sítí. Na vytvořených dvourozměrných geometrických modelech, reprezentujících tyto náhodné puklinové sítě, byly provedeny numerické simulace mechanického chování hornin při různých zátěžových podmínkách. Simulace provedené v simulátoru Flow123d poskytly tenzorová pole napětí a deformace. Následovala analýza těchto výsledků za účelem stanovení efektivních elastických parametrů. Na závěr byla aplikována stochastická homogenizace, která spočívala v průměrování výsledků z více náhodných puklinových sítí vygenerovaných na základě stejného statistického rozdělení.

Klíčová slova: diskrétní puklinová síť, efektivní elastický tenzor, numerická a stochastická homogenizace, nelineární mechanika krystalických hornin, deformace

Homogenization in rock mechanics

Abstract

This bachelor thesis focuses on the numerical homogenization of the mechanical properties of crystalline rocks, whose heterogeneous structure is characterized by a discrete fracture network. Numerical simulations of the mechanical behaviour of these rocks under various loading conditions were conducted using two-dimensional geometric models developed to represent the random fracture networks. The simulations, carried out with the Flow123d software, provided tensor fields of stress and deformation. The results were analysed to determine effective elastic parameters. Finally, stochastic homogenization was performed by averaging the results from multiple random fracture networks generated based on the same statistical distribution.

Keywords: discrete fracture network, effective elastic tensor, numerical and stochastic homogenization, nonlinear mechanics of crystalline rocks, deformation

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List of abbreviations

BGEM	B-spline GEometry Modeling package (SW library)
DFM	discrete fracture matrix (model)
DFN	discrete fracture network
FFT	fast Fourier transform
FEM	finite element method
FVM	finite volume method
GMSH	open source 3D finite element mesh generator (SW programme)
REV	representative elementary volume
SKB	Svensk Kärnbränslehantering AB (Swedish State Organisation for Radioactive Waste Management)
SW	software

List of used operators

$\sim$	Voigt notation
$\cdot$	semidiscretization
$\langle \cdot \rangle$	averaging operator for homogenization
$\{ \{ \cdot \} \}$	averaging operator for semidiscretization
∇, div	gradient, divergence
$\overline{\nabla}, \nabla_{\tau}, \nabla_{\nu}$	semidiscretized gradient, tangential, normal gradient
$\overline{\text{div}}, \text{div}_{\tau}, \text{div}_{\nu}$	semidiscretized divergence, tangent, normal divergence

List of Latin symbols used

a_f	fracture opening depending on the displacement field
$\mathbf{a}$	unit mean normal vector of fractures
$\mathbf{C}, \mathbf{C}_m, \mathbf{C}_f$	Hook elastic tensor, elastic tensor for the area Ω_m , elastic tensor for the area Ω_f
$\tilde{\mathbf{C}}^*$	effective elastic tensor in Voigt notation
$\mathbf{E}$	general load matrix
$\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3$	linearly independent load matrices for the respective boundary conditions
$\mathbf{f}, \mathbf{f}_m, \mathbf{f}_f$	external applied (volumetric) load force in $\Omega, \Omega_m, \Omega_f$
$F(\phi), G(r)$	distribution functions for the orientation and size of fractures
$g(r)$	probability density for the size (diameter) of the fractures
g_0	normalisation constant for $g(r)$
$\mathbf{I}$	second order unit tensor
k_r	fracture size scaling exponent
N	total number of fractures

p, p_m, p_f	pressure field, pressure in matrix, pressure in fractures
P_{30}	fracture intensity, i.e. the number of fractures per examined volume
P_{32}	areal intensity of fractures, i.e. the total area of fractures in relation to the examined volume
r	size i.e. diameter of fractures
$[r_0, r_\infty]$	size range of natural fractures
$[r_{\min}, r_{\max}]$	size range for artificial fractures
S	unit sphere
$\mathbf{t}_N$	specified surface force
$\mathbf{u}$	displacement vector field
$\mathbf{u}_D$	specified displacement vector field
$\mathbf{u}_f, \mathbf{u}_m$	displacement vector field in fractures, displacement vector field in matrix
$\mathbf{u}_{D1}, \mathbf{u}_{D2}, \mathbf{u}_{D3}$	vector displacement fields for each boundary condition
V	examined volume
$\mathbf{x}$	vector determining the position of the point before deformation
$\mathbf{y}$	vector determining the position of the point after deformation
Y, Y_m, Y_f	Young's modulus, Young's modulus of rock matrix, Young's modulus of fractures

List of used Greek symbols

β	load parameter i.e. relative magnitude of displacement
$\delta, \delta_{\min}$	počáteční rozevření pukliny před deformací, nezáporný parametr určující nejmenší povolenou velikost pukliny
η	the ratio of the fracture opening to its length
Γ_{Nm}, Γ_{Dm}	two disjunctive parts $\partial\Omega_m$
Γ_{Nf}, Γ_{Df}	two disjunctive parts $\partial\Omega_f$
κ_f, κ_v	concentration parameter for Fisher distribution, concentration parameter for Von Mises distribution
$\mathbf{\Lambda}^\pm, \mathbf{\Lambda}_\tau$	contact force on opposite sides of the fracture, contact force in tangential direction
ν	Poisson's number
$\Omega, \Omega_m, \Omega_f$	investigated continuous open set i.e. computational area, rock matrix area, fracture area
$\partial\Omega, \partial\Omega_m, \partial\Omega_f$	boundaries of the investigated continuous open set i.e. boundaries of the computational domain, boundaries of the rock matrix domain, boundaries of the fracture domain
$\boldsymbol{\nu}, \boldsymbol{\tau}$	normal vector to the fracture, tangent vector to the fracture
$\phi_d, \phi_s, \phi_t, \phi_p$	angle dip, strike, trend, plunge
ϕ_{tf}, ϕ_{tv}	angle trend for the Fisher distribution, the trend for the Von Mises distribution

$\boldsymbol{\sigma}, \boldsymbol{\sigma}_m, \boldsymbol{\sigma}_f$	stress tensor, stress tensor in matrix, stress tensor in fractures
$\boldsymbol{\varepsilon}$	small deformation tensor
$\bar{\boldsymbol{\varepsilon}}$	semidiscretized small deformation tensor
$\langle \tilde{\boldsymbol{\sigma}} \rangle, \langle \tilde{\boldsymbol{\varepsilon}} \rangle$	effective stress tensor in Voigt notation, effective small deformation tensor in Voigt notation
ξ	ratio of Young's modulus of fractures and rock matrix

List of other symbols

$\cdot$	dot product of vectors
$:$	dot product of second order tensors
$\otimes$	tensor product of vectors
$\emptyset$	empty set
∂	set boundary
$\text{cl}()$	set closure
$\top$	transposition
$()^+, ()^-$	value on opposite sides of the fracture
$\mathbb{N}$	natural numbers
$\mathbb{R}, \mathbb{R}^+, \mathbb{R}_0^+$	all real numbers, positive real numbers, positive real numbers including zero

1 Introduction

We have all encountered a material that has a heterogeneous, i.e. spatially variable, microstructure. There are a large number of materials with this microstructure, but some of the most important ones are crystalline rocks, biomaterials, wood and composite materials. What do all these materials have in common? Because of their very complex microstructure, it is extremely difficult to predict or model their mechanical properties in general. Mechanical properties such as stiffness, strength or elasticity are characteristics of a material that describe its behaviour and response to different mechanical loads. These properties are absolutely central to a deeper understanding of the behaviour of a material, because if we do not know the mechanical properties of a material, it is difficult to predict its behaviour in different situations. In other words, general modelling of heterogeneous materials with complex microstructure is a very challenging to almost impossible task, but one that must be addressed.

Numerical homogenization offers one possible way to approach this problem. This bachelor thesis focuses specifically on the application of numerical homogenization to the modeling of crystalline rocks containing a random fracture network. The importance of this topic lies, among other things, in the fact that crystalline rocks form a significant part of the Earth's crust and are undoubtedly an important material related to, for example, nuclear waste repositories or geothermal plants.

This issue will first be formulated theoretically. We will focus on the mathematical description of fissures, especially on the description of their orientation, distribution and size. We will then discuss the nonlinear mechanics of the fissure network, focusing, among other things, on the definition of the boundary and contact conditions of our problem.

The next part of the paper is devoted to the implemented program itself and its different parts. Initially, the basic scheme of the chosen procedure and the methods used for the generation of fissures and fissure networks are described. These networks are generated using the BGEM package, which allows the generation of fissures that are then grouped into families in various ways according to statistical distributions, orientations, lengths and other parameters. Once the fracture network has been generated, the study area is discretized using a two-dimensional finite element mesh, which is optimized before simulations are performed in the Flow123d simulator. These simulations result in tensor stress and strain fields.

Finally stochastic homogenization will be applied, which consists in averaging the results from multiple randomly generated fracture networks with the same statistical distribution. In addition, the influence of each parameter is analyzed.

2 Theory

The theoretical part is structured into four subsections. First, Chapter 2.1 summarizes the fundamental concepts from the mathematical theory of the mechanics of elastic bodies. Then, Chapter 2.2 describes the relationships determining the size, orientation, and distribution of fractures in crystalline rock. Chapter 2.3 subsequently introduces the model of nonlinear mechanics of a discrete fracture network, including boundary and contact conditions. The final Chapter 2.4 is dedicated to numerical homogenization and the general procedure for obtaining the effective elastic tensor.

2.1 Mathematical description of mechanics of elastic bodies

We start by introducing the necessary terms and markings. Dot product of vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ is defined as follows:

$$\mathbf{a} \cdot \mathbf{b} = \sum_{i=1}^n a_i b_i. \quad (2.1)$$

Similarly, the scalar product of the second order tensors $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$ is defined as follows:

$$\mathbf{A} : \mathbf{B} = \sum_{i=1}^n \sum_{j=1}^n A_{ij} B_{ij}. \quad (2.2)$$

Furthermore, the tensor product¹ of the vectors $\mathbf{v} \in \mathbb{R}^n$ and $\mathbf{w} \in \mathbb{R}^m$ is a second order tensor:

$$\mathbf{v} \otimes \mathbf{w} = \begin{bmatrix} v_1 w_1 & \cdots & v_1 w_m \\ v_2 w_1 & \cdots & v_2 w_m \\ \vdots & \ddots & \vdots \\ v_n w_1 & \cdots & v_n w_m \end{bmatrix} \in \mathbb{R}^{n \times m}. \quad (2.3)$$

Closure² of an open set $\mathcal{O}$ is defined as the union of its interior and boundary:

$$\text{cl}(\mathcal{O}) = \mathcal{O} \cup \partial\mathcal{O}. \quad (2.4)$$

¹Also known as exterior product

²More common is $\overline{\mathcal{O}}$. Here, cl was chosen to avoid a collision of markings in chap. 2.3.2.

The matrix $\mathbf{A}^\top$ denotes the transpose matrix, which is formed from the matrix $\mathbf{A}$ by mutual exchange of rows and columns, for elements valid:

$$(\mathbf{A}^\top)_{ij} = (\mathbf{A})_{ji}, \quad \forall i, j. \quad (2.5)$$

A gradient is a vector that is defined by the action of the nabla operator on a scalar field ($F : \mathbb{R}^n \rightarrow \mathbb{R}$) or vector field ($\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^n$):

$$\nabla F = \left[\frac{\partial F}{\partial x_1}, \frac{\partial F}{\partial x_2}, \dots, \frac{\partial F}{\partial x_n} \right]^\top, \quad \text{resp.} \quad \nabla \mathbf{F} = \left[\frac{\partial \mathbf{F}}{\partial x_1}, \frac{\partial \mathbf{F}}{\partial x_2}, \dots, \frac{\partial \mathbf{F}}{\partial x_n} \right]^\top. \quad (2.6)$$

For the vector field $\mathbf{F} = (F_1, F_2, \dots, F_n)$ the divergence is defined as a scalar:

$$\text{div } \mathbf{F} = \frac{\partial F_1}{\partial x_1} + \frac{\partial F_2}{\partial x_2} + \dots + \frac{\partial F_n}{\partial x_n}. \quad (2.7)$$

2.1.1 Deformation and displacement

Our aim is to describe the response of a heterogeneous body or material under load or **during deformation**. In this work, we consider two-dimensional (plane) problems for simplicity.

The term deformation of a body means a change in the shape or volume of a body caused by the action of some force. These forces, also known as deformation forces, cause stresses in the body and there are several types of them. The most common deformation forces are tensile, compressive, shear, bending or torsion. As such, deformations are divided into two basic types, namely **elastic** and **plastic** (inelastic). Elastic deformation is a reversible process, i.e. if a body is elastically deformed, then after the action is over the material returns to its original state. If it is plastic deformation, we speak of irreversible processes. In other words, in plastic deformation, the shape of the material is permanently changed (Brdička et al., 2005, chap. 5.2.). It is quite natural to conclude that elastic deformations will only occur for relatively small forces (depending on the type of material, of course), while plastic deformations will occur for larger forces. For the mathematical description of elastic deformation, we first introduce the so-called **vector displacement field**. Let us have an arbitrary point whose Cartesian coordinates before deformation we denote by x_i . The external forces will cause the point described by the vector field $\mathbf{u}(\mathbf{x})$ to shift to another point with coordinates y_i . The displacement vector field (or displacement field for short) $\mathbf{u}(\mathbf{x})$ is then defined as follows (Brdička et al., 2005, chap. 4.1.):

$$\mathbf{u}(\mathbf{x}) = \mathbf{y} - \mathbf{x}, \quad (2.8)$$

where vector $\mathbf{x}$ with components $x_i, i = 1, 2$ is the position of the point before deformation and vector $\mathbf{y}$ with components $y_i, i = 1, 2$ is the position of the point after deformation.

It must be emphasized that the displacement vector field is a **generic** description of the change in position of a point. In other words, while displacement fields are related to deformation, they are not equivalent concepts. For example, translation

(physical displacement) or rotation will lead to a non-trivial (non-zero) displacement field, i.e. $\mathbf{u}(\mathbf{x}) \neq 0$. However, there is no deformation during these motions because there is no change in the shape and volume of the body. The change in shape is described by the so-called **tensor of small deformations**(Zohdi and Wriggers, 2005, chap. 2.1.2):

$$\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2} \left(\nabla \mathbf{u} + (\nabla \mathbf{u})^\top \right). \quad (2.9)$$

2.1.2 Generalized Hooke's law, balance of forces and boundary conditions

The linear elastic material is described by **generalized Hooke's law**:

$$\boldsymbol{\sigma} = \mathbf{C}\boldsymbol{\varepsilon}(\mathbf{u}), \quad (2.10)$$

where $\boldsymbol{\sigma}$ is the stress tensor and $\mathbf{C}$ is Hooke's fourth order elastic tensor. For an isotropic material, this tensor can be represented by two constants - Young's modulus $Y > 0$ and Poisson's number³ ν :

$$\mathbf{C}\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{Y}{1+\nu}\boldsymbol{\varepsilon}(\mathbf{u}) + \frac{Y\nu}{(1+\nu)(1-2\nu)}(\operatorname{div} \mathbf{u})\mathbf{I}, \quad (2.11)$$

where $\mathbf{I}$ is the second order unit tensor. The balance of forces in this material is expressed by the system of Lamé equations, which can be written in the form:

$$\operatorname{div} \boldsymbol{\sigma} = \mathbf{f}, \quad (2.12)$$

where $\mathbf{f}$ is the applied volumetric force.

The interaction of the body with the environment is described by boundary conditions. Two basic types are most commonly used. The **Dirichlet** condition prescribes the displacement on the surface of the solid:

$$\mathbf{u} = \mathbf{u}_D, \quad (2.13)$$

while the **Neumann** condition gives the external surface force:

$$\boldsymbol{\sigma}\mathbf{n} = \mathbf{t}_N. \quad (2.14)$$

Here $\mathbf{u}_D$ represents the specified displacement field, $\mathbf{n}$ is the external unit normal vector and $\mathbf{t}_N$ is the specified surface force.

³Poisson's number is defined as the negative ratio of the longitudinal and transverse strain and for most materials lies in the interval $(0, \frac{1}{2})$. However, there are also so-called *auxetic* materials with $\nu < 0$.

2.1.3 Voigt notation

In mathematics, Voigt notation is a way to represent a **symmetric** tensor by a lower order tensor (Povey, 2023).

Consider the second order two-dimensional symmetric tensor $\mathbf{A} \in \mathbb{R}^{2 \times 2}$ and the fourth order two-dimensional symmetric tensor $\mathbf{B} \in \mathbb{R}^{2 \times 2 \times 2 \times 2}$.

In Voigt notation, the tensor $\mathbf{A}$ reduces to a first order tensor (vector) and the tensor $\mathbf{B}$ reduces to a second order tensor (matrix):

$$\begin{aligned} \mathbf{A} \in \mathbb{R}^{2 \times 2} &\longrightarrow \tilde{\mathbf{A}} \in \mathbb{R}^3, \\ \mathbf{B} \in \mathbb{R}^{2 \times 2 \times 2 \times 2} &\longrightarrow \tilde{\mathbf{B}} \in \mathbb{R}^{3 \times 3}, \end{aligned} \quad (2.15)$$

where $\tilde{\mathbf{A}}$ and $\tilde{\mathbf{B}}$ denote the reduced tensors. These reduced tensors contain only *various* elements of the original tensors:

$$\tilde{\mathbf{A}} = \begin{bmatrix} A_{11} \\ A_{22} \\ A_{12} \end{bmatrix}, \quad \tilde{\mathbf{B}} = \begin{bmatrix} B_{1111} & B_{1122} & B_{1112} \\ B_{2211} & B_{2222} & B_{2212} \\ B_{1211} & B_{1222} & B_{1212} \end{bmatrix}.$$

2.2 Mathematical description of fractures

Crystalline rocks are often characterized by the presence of fractures, which are linear and planar formations whose properties differ **significantly** from the surrounding rock. These fractures are usually the result of various geological processes and have a major influence on the hydromechanical (and other related) processes in the rock mass. The fracture network is usually random in nature and its structure is described by several basic quantities.

The first key property of the fracture network is **orientation**. The orientation of the fracture network defines the direction in which the fractures lie in space and is usually described by different angles and probability densities.

Another important quantity for describing fractures in a fracture network is **size**, which is usually defined as the length of linear fractures or the area of planar fractures. This parameter directly determines the effect of a given fracture on the surrounding environment, e.g. fracture networks with larger fractures usually exhibit lower strength, which will cause greater resulting deformation.

The **Intensity** of the fracture network indicates how many fractures are present in a given space or plane. It can be defined as either the areal density or the number of fractures per unit volume of rock. The intensity is also closely related to the **location** or **spatial distribution** of the fractures.

All these quantities can be determined to some extent from experimental data, namely from outcrops, boreholes or tunnels. The process of extrapolating these quantities from experimental data is very complicated, as it involves not only mechanical but also hydrological aspects and is usually performed in three dimensions.

For this reason, we only refer to relevant articles published in *Svensk Kärnbränslehantering AB* (SKB), where this issue is discussed in more detail and in

general. These include e.g.: (Oehman and Sven Follin, 2010, chap. 2 a Appendix E, F, H, I), (S. Follin et al., 2006, chap. 2, 3) or alternatively (Davis, Sampson, et al., 2002, chap. 4, 5).

2.2.1 Orientation of fractures

Several geological parameters can be used to describe the orientation of each fracture in the studied area.

The angles *dip*⁴ and *strike* are most often used to describe the orientation of a planar fissure in space. The first of these two angles is denoted ϕ_d and expresses the steepest angle of descent of the fracture plane. The strike angle, ϕ_s , is the angle between the line given by the intersection of the fracture plane with the horizontal plane and the north (Maptek Pty Ltd, 2024).

The angles *trend* and *plunge* are used to describe the orientation of a linear fracture in space. The trend, ϕ_t , is the angle between the north and the projection of the fracture in the horizontal plane. Plunge, ϕ_p , is the angle between the fracture and the horizontal plane. The angles ϕ_d and ϕ_p are defined on the interval $(0, \frac{\pi}{2})$, the angles ϕ_t and ϕ_s lie in the interval $(0, 2\pi)$. These angles are then used as one of several input parameters in the generation of the fracture network (see Ch. 3.2.1), where it is also necessary to distinguish between the ϕ_t angles for the Fisher distribution, where the trend angle is referred to as ϕ_{tf} , and the Von Mises distribution, where the trend angle is introduced as ϕ_{tv} .

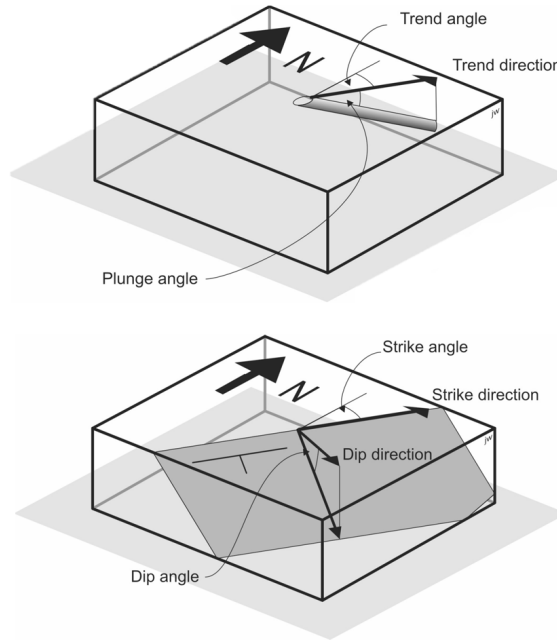


Figure 2.1: Visualization of angles and directions determining the orientation of fractures, taken from (Waldron and Snyder, 2020, str. 21/22)

⁴For all the following parameters there is no unambiguous English translation, so they have been left in English form.

In the picture 2.1 we can see the visualization of all mentioned parameters. In the upper part of the figure we have the linear fracture and the corresponding trend, plunge, trend direction and north angles. In the lower part we have a planar fracture and the corresponding strike, dip, vectors determining the direction of the plane (strike, dip direction) and north.

2.2.2 Probability distribution for fracture orientation

As mentioned in the introduction of the chapter 2.2, the fractures show a random character. That is, while individual fractures are randomly oriented, their orientations are assumed to be centered around a certain direction given by the unit vector $\mathbf{a}$.

For a probabilistic description of the orientation in p -dimensional space, the *Von Mises-Fisher* distribution is generally used (see for details (Figueiredo, 2009, Ch. 2)). The following two subsections are therefore only a particular case of this generalized distribution, in three-dimensional and two-dimensional space.

Fisher distribution in 3D

To describe orientation in 3D, a one-dimensional Fisher distribution is used, whose probability density is defined as follows:

$$f_3(\mathbf{n} \mid \mathbf{a}, \kappa_f) = \frac{\kappa_f e^{\kappa_f (\mathbf{n} \cdot \mathbf{a})}}{2\pi (e^{\kappa_f} - e^{-\kappa_f})}, \quad (2.16)$$

where $\mathbf{n}$ is the unit normal vector of a random fracture, $\mathbf{a}$ is the unit mean normal vector of the fractures, and $\kappa_f > 0$ is a parameter called *Fisher concentration* determining the dispersion of the fractures, whose physical interpretation will be discussed in more detail later. If we denote by $\theta \in [0, \pi]$ the angle between the vectors $\mathbf{n}$ and $\mathbf{a}$, then the following holds for the Fisher distribution function:

$$\int_S f_3(\mathbf{n} \mid \mathbf{a}, \kappa_f) d\mathbf{n} = \int_0^\pi f(\theta) d\theta, \quad (2.17)$$

where S is the unit sphere and $f(\theta)$ holds:

$$f(\theta) = \frac{\kappa_f e^{\kappa_f \cos \theta} \sin \theta}{e^{\kappa_f} - e^{-\kappa_f}}. \quad (2.18)$$

If we integrate $f(\theta)$ we get a new distribution function $F(\theta)$:

$$F(\theta) = \int_0^\theta f(\vartheta) d\vartheta = \frac{e^{\kappa_f} - e^{\kappa_f \cos \theta}}{e^{\kappa_f} - e^{-\kappa_f}}. \quad (2.19)$$

It is now possible to analytically express θ from the equation (2.19):

$$\theta = \cos^{-1} \left\{ \frac{1}{\kappa_f} \ln (e^{\kappa_f} - F(\theta) (e^{\kappa_f} - e^{-\kappa_f})) \right\}. \quad (2.20)$$

The equation (2.20) allows to generate random orientations of θ from a uniform distribution of values of the distribution function F . For the limiting case $\kappa_f \rightarrow 0$ we get a *uniform* distribution, i.e. the normals, and hence the fractures, can have any direction. In this case, θ is determined as follows:

$$\theta = \cos^{-1}(1 - 2F), \quad (2.21)$$

while for the limit $\kappa_f \rightarrow \infty$ we have no dispersion between the fractures. In other words, all normals are exactly in the direction defined by $\mathbf{a}$ and are therefore parallel to each other. As far as the distribution is concerned, the case of $\kappa_f \rightarrow \infty$ corresponds to the *Dirac delta* distribution⁵.

Von Mises' division in 2D

The Von Mises distribution can be used for the angle of rotation of the fracture around the normal or the orientation of fractures in 2D. Its probability density is given by the relation:

$$f_2(\mathbf{n} \mid \mathbf{a}, \kappa_v) = \frac{e^{\kappa_v(\mathbf{n} \cdot \mathbf{a})}}{2\pi I_0(\kappa_v)}, \quad (2.22)$$

where

$$I_0(\kappa_v) = \frac{1}{\pi} \int_0^\pi e^{\kappa_v \cos \theta} d\theta. \quad (2.23)$$

For increasing κ_v , the Von Mises distribution approximates the *normal*⁶ distribution with mean $\mathbf{a}$ and variance $\frac{1}{\kappa_v}$.

2.2.3 Size of natural fractures

In geology, the size of fractures can be studied at different scales - from centimetres to several metres at the surface of outcrops, or even in larger structures such as fault zones. Fault zones are elongated clusters of fractures that range in size from hundreds of metres to several kilometres (S. Follin et al., 2006, chap. 2.1).

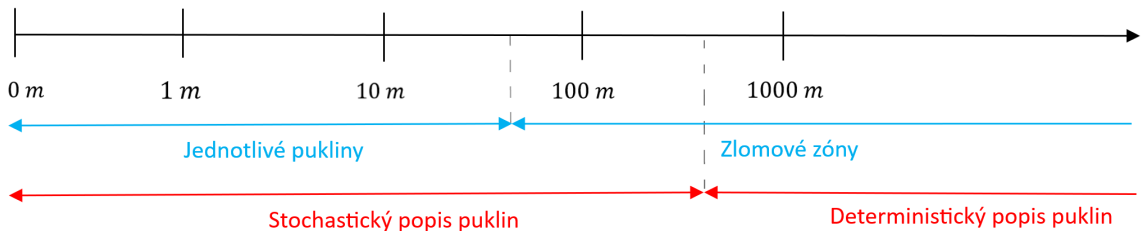


Figure 2.2: Representation of the relationship between fault zones (deterministic description) and the fracture network (stochastic description), inspiration (S. Follin et al., 2006, fig. 2.1).

⁵We will not discuss in detail here, for more details see (Hur, 2009).

⁶Also called Gaussian distribution.

Related to this is a key parameter that characterizes the size distribution of fractures, the so-called **fracture size scaling exponent** $k_r > 2$.

This parameter describes the probability distribution of the fracture sizes in a given area. The probability density for the size (average) of the r fractures is usually described by a truncated power law:

$$g(r) = \frac{1}{g_0} r^{-(k_r+1)}, \quad (2.24)$$

where the normalisation constant

$$g_0 = \int_{r_0}^{r_\infty} r^{-(k_r+1)} dr = \frac{r_0^{-k_r} - r_\infty^{-k_r}}{k_r} \quad (2.25)$$

is chosen with respect to the range of natural fracture sizes $[r_0, r_\infty]$. The distribution function $G(r)$ is given by:

$$G(r) = \int_{r_0}^r g(\mu) d\mu = \frac{k_r}{r_0^{-k_r} - r_\infty^{-k_r}} \int_{r_0}^r \mu^{-(k_r+1)} d\mu = \frac{r_0^{-k_r} - r^{-k_r}}{r_0^{-k_r} - r_\infty^{-k_r}}, \quad (2.26)$$

whose inverse gives the expression for r :

$$r = \left\{ r_0^{-k_r} - G(r)(r_0^{-k_r} - r_\infty^{-k_r}) \right\}^{-\frac{1}{k_r}}. \quad (2.27)$$

2.2.4 Intensity and spatial distribution of natural fractures

The factor that has the greatest influence on the amount or intensity of fractures formed in a given interval is the number of large fractures in the rock volume. However, the total number of fractures is the result of a complex interplay of various parameters describing the fractures. For a given intensity, the model may contain either few large fractures or a large number of much smaller fractures (S. Follin et al., 2006, chap. 3.1).

For the description of the intensity of the fractures, a designation is introduced using two numbers in the index of the respective quantity, which express the respective intensity dimensions. For example, $P_{32}[r_0, r_\infty]$ denotes the total area of fractures relative to the volume under study. The unit $P_{32}[r_0, r_\infty]$ is $\left[\frac{m^2}{m^3}\right]$. Similarly, $P_{30}[r_0, r_\infty]$ can be defined with the unit $\left[\frac{1}{m^3}\right]$:

$$P_{30}[r_0, r_\infty] = \frac{N}{V}, \quad (2.28)$$

where $N \in \mathbb{N}$ is the total number of fractures and $V > 0$ is the volume under study. The equation (2.28) can also be written equivalently as follows (Munier, 2007, Chap. 3.1):

$$P_{30}[r_0, r_\infty] = \frac{P_{32}[r_0, r_\infty]}{I}, \quad I = \int_{r_0}^{r_\infty} \pi r^2 g(r) dr, \quad (2.29)$$

where:

$$I = \frac{\pi k_r}{r_0^{-k_r} - r_\infty^{-k_r}} \int_{r_0}^{r_\infty} r^{1-k_r} dr = \frac{\pi k_r (r_0^{2-k_r} - r_\infty^{2-k_r})}{(r_0^{-k_r} - r_\infty^{-k_r}) (k_r - 2)}. \quad (2.30)$$

The equation (2.30) is defined for $k_r > 2$.

2.2.5 Size and intensity of artificially generated fractures

The exact geometry of the natural fracture network is not usually known. For computational purposes, an artificially generated fracture network is used, which has the same probabilistic characterization as the natural network.

In order to make the number of generated fractures both representative and reasonable, it is necessary to restrict the range of their sizes to a smaller interval $[r_{\min}, r_{\max}] \subseteq [r_0, r_{\infty}]$, see Fig. 2.3. For the intensity of fractures on this subinterval:

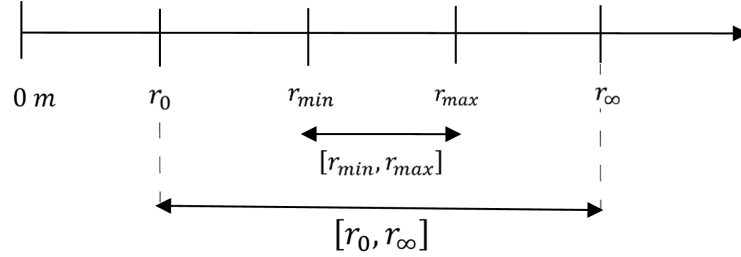


Figure 2.3: Visualisation of the intervals used to generate the artificial fracture network

$$P_{30}[r_{\min}, r_{\max}] = P_{30}[r_0, r_{\infty}] \int_{r_{\min}}^{r_{\max}} g(r) dr. \quad (2.31)$$

If we add the previous dependencies, i.e. (2.30), (2.29), to the relation (2.31) and determine the value of the integral, we get the final form for $P_{30}[r_{\min}, r_{\max}]$:

$$P_{30}[r_{\min}, r_{\max}] = \frac{P_{32}[r_0, r_{\infty}] (k_r - 2) (r_{\min}^{-k_r} - r_{\max}^{-k_r})}{\pi k_r (r_0^{2-k_r} - r_{\infty}^{2-k_r})}. \quad (2.32)$$

For $k_r = 3$ the probabilistic fracture model is *self similar*, i.e. the probability distributions of the relevant quantities are independent of scale (area size).

2.3 Mathematical model for rock mechanics

In the previous subsection, we discussed the mathematical description of a single fracture or fractures. In other words, we have not dealt at all with how to mathematically describe the interaction of a fracture network with its surroundings and with itself, which is undoubtedly very important for understanding the model we are using of the combination of discrete fractures and matrix. The mathematical description of this model in terms of mechanics is the aim of this subsection.

The first in this chapter is a general introduction, where we introduce the different rock models with fractures, how they differ from each other and how they relate to the context of this thesis. We then focus on an introduction to the mathematical description of a model combining a discrete fracture network and a rock matrix and

the necessary definitions describing our model. We first discuss the semidiscretization of differential operators and various quantities. Finally, these operators (and more) will be used to formulate the contact and boundary conditions, which are crucial for our model. In the meantime, we discuss the general mechanics combining matrix and discrete fracture networks.

2.3.1 General introduction to the issue

There are currently three types of approaches to describing fractures in continuum-based models:

- continuum equivalent models,
- discrete fracture network models, denoted DFN - discrete fracture network,
- discrete fracture matrix models, denoted DFM - discrete fracture matrix.

The first two models mentioned here will not be discussed in detail here⁷, but we will at least introduce them very briefly to better understand the concept of DFM.

In equivalent continuum models, the influence of small fractures is included in the macroscopic properties of the rock, so they are also suitable for describing rocks containing many small fractures. However, these models do not capture the very high ratio of the fracture size to the fracture width, which leads to the neglect of large flow velocity and displacement gradients that cannot be addressed by classical discretization methods (such as the finite volume method, FVM, or finite element method, FEM).

In contrast, the DFN models describe individual fractures directly. These models are suitable for the detailed description of specific fractures and we do not have the problem of gradients, but on the other hand they neglect the properties of the rock matrix and are not able to effectively describe the problem with a large number of small fractures.

Then there are various hybrid models, which include DFM models. These models describe the physical processes not only in the rock matrix but also in the fracture network and their interactions. Usually DFM models are also associated with flow (hydromechanical models), but this is beyond the scope of this paper.

In the following subsections we will present a **simplified DFM** model describing **mechanics**. The full hydromechanical model is derived and described in detail in (Stebel et al., 2024) or (Březina and Stebel, 2024).

2.3.2 Semidiscretization

Let Ω be a continuous open set (region) in $\mathbb{R}^2$ with a Lipschitz⁸ boundary $\partial\Omega$. We further assume that Ω is composed of two subregions:

⁷For more details and context, see (Berre et al., 2019)

⁸For the definition of the Lipschitz boundary, see (Nečas, 2011, Ch. 1.1.3).

The first subregion, Ω_m , is a **matrix** whose boundary $\partial\Omega_m$ is composed of two disjunctive parts, i.e. $\partial\Omega_m = \Gamma_{Nm} \cup \Gamma_{Dm}$, where $\Gamma_{Nm} \cap \Gamma_{Dm} = \emptyset$. The second subregion, Ω_f , denotes the **discrete fracture network** and for its boundary $\partial\Omega_f = \Gamma_{Nf} \cup \Gamma_{Df}$, where $\Gamma_{Nf} \cap \Gamma_{Df} = \emptyset$. We assume that Ω_f is the union of the segments whose tangent and normal vectors will be denoted by $\boldsymbol{\tau}$ and $\boldsymbol{\nu}$, respectively. The width of the fracture represented by a given line segment will be denoted by the symbol δ .

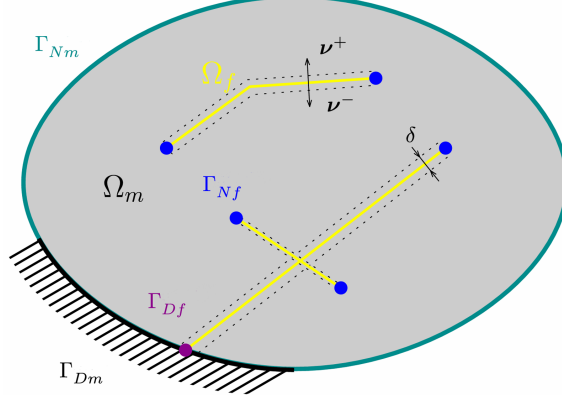


Figure 2.4: Visualization of the DFM mechanics, inspiration (Stebel et al., 2024, fig. 1)

In Ω_f we introduce so-called **semidiscretized operators**, which will be **marked with a bar**. Verbally, the semidiscretized operators and quantities could be described as certain *approximations* of the full versions of these operators and quantities.

In general, the relation for a semidiscrete gradient of a scalar (e.g. pressure field $p = (p_m, p_f)$, where p_m, p_f denote the pressure in the matrix and in the fractures, respectively) and vector field (e.g. The displacement field $\mathbf{u} = (\mathbf{u}_m, \mathbf{u}_f)$) should be defined as follows⁹:

$$\begin{aligned}\nabla p &\approx \overline{\nabla} p = \nabla_{\boldsymbol{\tau}} p_f + \nabla_{\boldsymbol{\nu}} p, \\ \nabla \mathbf{u} &\approx \overline{\nabla} \mathbf{u} = \nabla_{\boldsymbol{\tau}} \mathbf{u}_f + \nabla_{\boldsymbol{\nu}} \mathbf{u},\end{aligned}\tag{2.33}$$

where $\nabla_{\boldsymbol{\tau}}, \nabla_{\boldsymbol{\nu}}$ denote tangent and normal gradients, respectively. Next, we define the semidiscrete divergence for the vector field:

$$\operatorname{div} \mathbf{u} \approx \overline{\operatorname{div}} \mathbf{u} = \operatorname{div}_{\boldsymbol{\tau}} \mathbf{u}_f + \operatorname{div}_{\boldsymbol{\nu}} \mathbf{u},\tag{2.34}$$

where $\operatorname{div}_{\boldsymbol{\tau}}, \operatorname{div}_{\boldsymbol{\nu}}$ is tangent and normal divergence, respectively. Finally, let us also define the semidiscretization of the small deformation tensor, which was described in (2.9):

$$\overline{\boldsymbol{\varepsilon}}(\mathbf{u}) = \frac{1}{2} \left(\overline{\nabla} \mathbf{u} + (\overline{\nabla} \mathbf{u})^T \right).\tag{2.35}$$

⁹It would be more correct to write the dependence of the quantities on the space or on $\mathbf{x}$, but for the sake of space we will leave it like this

The differential operators in the normal direction expressed in the equations (2.33) and (2.34) can be described by *averaging operator* $\{\{\cdot\}\}$, which is defined as:

$$\nabla_{\nu} p = \{\{p\nu\}\} = \frac{\Delta^+ p \nu^+ + \Delta^- p \nu^-}{2}, \quad (2.36)$$

where $\Delta^{\pm} p = \frac{2}{\delta} (p_m^{\pm} - p_f)$ a $\nu^+ = \nu$ is the opposite vector to ν^- . In other words, the normal gradient of a scalar field can be interpreted as the averaged product of the value of this field and ν . By analogy for a vector field $\mathbf{u}$ we get:

$$\begin{aligned} \nabla_{\nu} \mathbf{u} &= \{\{\Delta \mathbf{u} \otimes \nu\}\} = \frac{\Delta^+ \mathbf{u} \otimes \nu^+ + \Delta^- \mathbf{u} \otimes \nu^-}{2}, \\ \text{div}_{\nu} \mathbf{u} &= \{\{\Delta \mathbf{u} \cdot \nu\}\} = \frac{\Delta^+ \mathbf{u} \cdot \nu^+ + \Delta^- \mathbf{u} \cdot \nu^-}{2}, \end{aligned} \quad (2.37)$$

where $\Delta^{\pm} \mathbf{u} = \frac{2}{\delta} (\mathbf{u}_m^{\pm} - \mathbf{u}_f)$. All these semidiscrete operators can also be defined for partial fracture boundaries, e.g. $\bar{\varepsilon}^{\pm}(\mathbf{u})$, $\bar{\text{div}}^{\pm} \mathbf{u}$. It's basically the same thing, but in the equations (2.36) and (2.37) instead of the averaging operator, the product of the field value and ν at the given fracture boundary is used directly. For illustrative purposes, we will show one example, namely $\bar{\varepsilon}^{\pm}(\mathbf{u})$:

$$\bar{\varepsilon}^{\pm}(\mathbf{u}) = \frac{1}{2} \left(\bar{\nabla}^{\pm} \mathbf{u} + \left(\bar{\nabla}^{\pm} \mathbf{u} \right)^T \right), \quad (2.38)$$

where $\bar{\nabla}^{\pm} \mathbf{u} = \nabla_{\tau} \mathbf{u}_f + \Delta^{\pm} \mathbf{u} \otimes \nu^{\pm}$.

2.3.3 DFM model description

The balance of forces in the DFM can be described as follows (Stebel et al., 2024, equations (2.6), (2.8), (2.10), (2.12)):

$$\left. \begin{aligned} -\text{div } \boldsymbol{\sigma}_m &= \mathbf{f}_m \\ \boldsymbol{\sigma}_m &= \mathbf{C}_m \boldsymbol{\varepsilon}(\mathbf{u}_m) \end{aligned} \right\} \text{v } \Omega_m, \quad \left. \begin{aligned} -\bar{\text{div}} (\delta \boldsymbol{\sigma}_f) &= \delta \mathbf{f}_f \\ \boldsymbol{\sigma}_f &= \mathbf{C}_f \bar{\varepsilon}(\mathbf{u}) \end{aligned} \right\} \text{in } \Omega_f, \quad (2.39)$$

where $\boldsymbol{\sigma}_{\star}$ is the stress tensor, $\mathbf{f}_{\star}$ is the external applied (load) force, $\boldsymbol{\varepsilon}$ is the small deformation tensor, $\mathbf{C}_{\star}$ is the Hooke elastic tensor, $\delta > 0$ is the initial fracture opening before deformation and $\star \in \{m, f\}$. The first equation in $\Omega_{\star}$ can be viewed as the balance of forces in $\Omega_{\star}$, see (2.12). The second equation is called **generalized Hooke's law**.

Since $\boldsymbol{\sigma}_{\star}$, $\boldsymbol{\varepsilon}$ and $\mathbf{C}_{\star}$ are symmetric tensors, they can be represented in the Voigt notation (2.15). However, a small correction is introduced for the small deformation tensor $\boldsymbol{\varepsilon}$:

$$\tilde{\boldsymbol{\varepsilon}} = \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \gamma_{12} \end{bmatrix}, \quad (2.40)$$

where $\gamma_{12} = 2\varepsilon_{12}$. This preserves the dot product $\tilde{\boldsymbol{\sigma}}_\star \cdot \tilde{\boldsymbol{\varepsilon}} = \boldsymbol{\sigma}_\star : \boldsymbol{\varepsilon}$, a quantity important for expressing mechanical energy. In Voigt notation, the generalised Hooke's law translates the equation (2.39) into the form:

$$\tilde{\boldsymbol{\sigma}}_m = \tilde{\mathbf{C}}_m \tilde{\boldsymbol{\varepsilon}}(\mathbf{u}_m) \quad \text{i.e.} \quad \tilde{\boldsymbol{\sigma}}_f = \tilde{\mathbf{C}}_f \tilde{\boldsymbol{\varepsilon}}(\mathbf{u}), \quad (2.41)$$

where

$$\tilde{\boldsymbol{\sigma}}_\star = \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{bmatrix}, \quad \tilde{\mathbf{C}}_\star = \begin{bmatrix} C_{1111} & C_{1122} & C_{1112} \\ C_{2211} & C_{2222} & C_{2212} \\ C_{1211} & C_{1222} & C_{1212} \end{bmatrix}, \quad (2.42)$$

which is a generalized Hooke's law in Voigt notation on subfields $\Omega_\star$.

2.3.4 Contact conditions

The transfer of forces between the rock matrix and the fracture is expressed by the so-called contact conditions. To express them, we first define the **contact force** acting on each fracture boundary:

$$\boldsymbol{\Lambda}^\pm = (\mathbf{C}_f^\pm \tilde{\boldsymbol{\varepsilon}}^\pm(\mathbf{u}) - \mathbf{C}_m \boldsymbol{\varepsilon}(\mathbf{u}_m^\pm)) \boldsymbol{\nu}^\pm. \quad (2.43)$$

The contact force acting in the tangential direction of the fracture will be denoted by $\boldsymbol{\Lambda}_\tau$. The last thing we need to define here is a width-related quantity, which has the following form¹⁰:

$$a_f(\mathbf{u}) = \delta + \mathbf{u}_m^+ \cdot \boldsymbol{\nu}^+ + \mathbf{u}_m^- \cdot \boldsymbol{\nu}^-. \quad (2.44)$$

Physically, the quantity $a_f(\mathbf{u})$ represents the width or the fracture opening depending on the displacement field around the fracture. The value of $a_f(\mathbf{u})$ is always positive, which corresponds to a *open fracture*. As soon as a force or displacement starts to act on the fracture, the fracture starts to close. Due to the roughness of the fracture surface and other effects, $a_f(\mathbf{u})$ cannot fall below a certain non-negative minimum value, which we call $\delta_{\min}$. This minimum value ensures that the fracture cannot penetrate through itself under the influence of the applied forces. This concept is part of one of the three contact conditions of DFM. Another of the contact conditions is the balance of contact forces, i.e. $\boldsymbol{\Lambda}^\pm$ must be opposite to each other. The last of these conditions is related to the fact that we neglect friction in the tangential direction of the contact forces at each fracture boundary and between the matrix. As a result, we get **complete contact conditions** (Stebel et al., 2024, equation (2.16) to (2.18)):

$$\begin{aligned} a_f(\mathbf{u}) &\geq \delta_{\min} && \text{(non-penetration).} \\ \boldsymbol{\Lambda}^+ &= -\boldsymbol{\Lambda}^- && \text{(balance of contact forces),} \\ \boldsymbol{\Lambda}_\tau^\pm &= \mathbf{0} && \text{(without friction).} \end{aligned} \quad (2.45)$$

For a better visualization of the contact mechanics, the following picture is suitable 2.5:

¹⁰Label taken from the English term *aperture*.

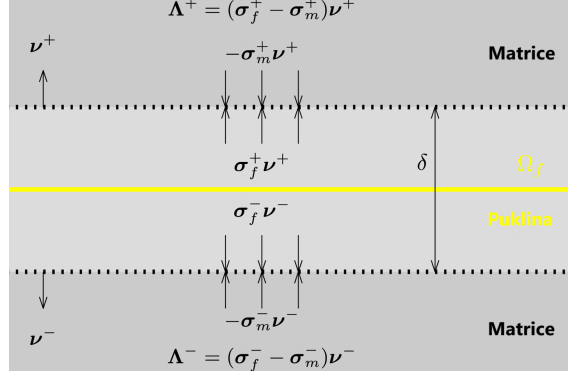


Figure 2.5: Visualization of contact mechanics, inspiration (Stebel et al., 2024, fig. 2)

2.3.5 Boundary conditions

For a complete description of the vector fields of displacement $\mathbf{u}$ and stress $\boldsymbol{\sigma}$ in the whole Ω region, it is also necessary to determine the **boundary conditions**:

$$\begin{aligned} \mathbf{u}_\star &= \mathbf{u}_{D\star} \quad \text{na } \Gamma_{N\star}, \\ \boldsymbol{\sigma}_\star \mathbf{n} &= \mathbf{t}_{N\star} \quad \text{na } \Gamma_{D\star}, \end{aligned} \quad (2.46)$$

where $\mathbf{n}$ is the unit external normal vector, $\mathbf{u}_D$ is the specified displacement vector field and $\mathbf{t}_N$ is the specified surface force.

2.4 Numerical homogenization

Numerical homogenization is a key tool for efficient modeling of heterogeneous materials, especially composites and porous structures. The main methods include the **representative elementary volume** (REV) method, also referred to as *upscaling*, which consists of calculating effective material properties based on a periodic or random sample of the microstructure (Hill, 1963). This often involves the use of FEM to solve auxiliary problems at the micro-level. A more complex interplay of micro and macroscopic phenomena is enabled by **multiscale homogenization**, in which problems at several scales are solved simultaneously (Kouznetsova et al., 2001). Another approach is the multidimensional **asymptotic homogenization**, which uses the development of unknown quantities according to a small parameter characterizing the ratio of scales (Bensoussan et al., 2011). Recently, numerical methods based on **fast Fourier transform** (FFT) have also been developed, which allow efficient computations on regular grids and are suitable for materials with complex microstructures (Moulinec and Suquet, 1998). In this paper, the representative elementary volume method is used to calculate the effective elastic tensor of a crystalline rock.

The goal of homogenization is to find equivalent parameters independent of $\mathbf{x}$ that account for the heterogeneity of the original variables. In particular, we want to replace the elastic tensor $\tilde{\mathbf{C}}_m(\mathbf{x})$ by a constant tensor $\tilde{\mathbf{C}}^*$.

2.4.1 Definition of homogenization and effective elastic tensor

The homogenization is performed using the averaging operation $\langle \cdot \rangle$, defined as:

$$\langle \cdot \rangle = \frac{1}{|\Omega_m|} \int_{\Omega_m} \cdot d\Omega_m. \quad (2.47)$$

This operation neglects the behavior of the relevant quantity in Ω_f , since the stiffness and stress are typically several orders of magnitude smaller in the fractures than in the rock matrix. If we apply this operator to the equation (2.41), we get:

$$\langle \tilde{\sigma} \rangle = \frac{1}{|\Omega_m|} \int_{\Omega_m} \tilde{\mathbf{C}}_m(\mathbf{x}) \tilde{\boldsymbol{\varepsilon}}(\mathbf{u}_m(\mathbf{x})) d\Omega_m. \quad (2.48)$$

Our goal is to replace the tensor $\tilde{\mathbf{C}}_m(\mathbf{x})$ with a tensor $\tilde{\mathbf{C}}^*$ such that it is independent of $\mathbf{x}$ and satisfies the following requirement:

$$\frac{1}{|\Omega_m|} \int_{\Omega_m} \tilde{\mathbf{C}}_m(\mathbf{x}) \tilde{\boldsymbol{\varepsilon}}(\mathbf{u}_m(\mathbf{x})) d\Omega_m = \tilde{\mathbf{C}}^* \frac{1}{|\Omega_m|} \int_{\Omega_m} \tilde{\boldsymbol{\varepsilon}}(\mathbf{u}_m(\mathbf{x})) d\Omega_m$$

i.e.

$$\langle \tilde{\sigma} \rangle = \langle \tilde{\mathbf{C}}_m \tilde{\boldsymbol{\varepsilon}} \rangle = \tilde{\mathbf{C}}^* \langle \tilde{\boldsymbol{\varepsilon}} \rangle. \quad (2.49)$$

The homogenization of the equation (2.41) can be written symbolically as follows:

$$\tilde{\sigma}_m = \tilde{\mathbf{C}}_m \tilde{\boldsymbol{\varepsilon}} \xrightarrow{\text{homogenization}} \langle \tilde{\sigma} \rangle = \tilde{\mathbf{C}}^* \langle \tilde{\boldsymbol{\varepsilon}} \rangle. \quad (2.50)$$

2.4.2 Problem boundary conditions and load matrix

Given the fact that $\tilde{\mathbf{C}}^* \in \mathbb{R}^{3 \times 3}$ and $\langle \tilde{\sigma} \rangle, \langle \tilde{\boldsymbol{\varepsilon}} \rangle \in \mathbb{R}^3$, the relation (2.50) is a system of 3 equations with 9 unknowns, which is not enough equations to find $\tilde{\mathbf{C}}^*$.

To completely determine all elements of the tensor $\tilde{\mathbf{C}}^*$ it is necessary to perform a number of **linearly independent** loads via **boundary conditions** defined as follows:

$$\mathbf{u}_D = \mathbf{E} \mathbf{x}, \quad (2.51)$$

where $\mathbf{u}_D$ is the displacement vector field, $\mathbf{E} = \begin{bmatrix} E_{11} & E_{12} \\ E_{12} & E_{22} \end{bmatrix}$ denotes the symmetric **general load matrix** and $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$.

By introducing these boundary conditions into the mechanics model (2.39), (2.46) we obtain the displacement field $\mathbf{u}$ and the stress field $\boldsymbol{\sigma}$ over the whole Ω region.

The result of averaging is then 3 *different averaged stress components*. Thus we get 3 equations for the elements of $\tilde{\mathbf{C}}^*$. In order to determine the entire tensor $\tilde{\mathbf{C}}^*$ we need 3 different cases to have 9 equations for 9 unknowns. In order to apply the

general load matrix to the equation *eqrefdefinitionoftheeffectiveelastictensor*, it is first necessary to define the relationship between $\tilde{\mathbf{E}}$ (the general load matrix in Voigt notation) and $\langle \tilde{\boldsymbol{\varepsilon}} \rangle$. This equality is a result of ‘*The Average Strain Theorem*’, which is described in detail in (Zohdi and Wriggers, 2005, kap. 4.1.1):

$$\tilde{\mathbf{E}} = \langle \tilde{\boldsymbol{\varepsilon}} \rangle \iff \begin{bmatrix} E_{11} \\ E_{22} \\ 2E_{12} \end{bmatrix} = \begin{bmatrix} \langle \varepsilon_{11} \rangle \\ \langle \varepsilon_{22} \rangle \\ \langle \gamma_{12} \rangle \end{bmatrix}. \quad (2.52)$$

We now show the procedure for determining the effective elastic tensor $\tilde{\mathbf{C}}^*$.

2.4.3 Effective elastic tensor for general load matrices

By applying three different load matrices, we obtain from the equation (2.49) conditions:

$$\begin{aligned} \langle \tilde{\boldsymbol{\sigma}} \rangle^1 &= \tilde{\mathbf{C}}^* \langle \tilde{\boldsymbol{\varepsilon}} \rangle^1, \\ \langle \tilde{\boldsymbol{\sigma}} \rangle^2 &= \tilde{\mathbf{C}}^* \langle \tilde{\boldsymbol{\varepsilon}} \rangle^2, \\ \langle \tilde{\boldsymbol{\sigma}} \rangle^3 &= \tilde{\mathbf{C}}^* \langle \tilde{\boldsymbol{\varepsilon}} \rangle^3, \end{aligned} \quad (2.53)$$

where $\langle \tilde{\boldsymbol{\sigma}} \rangle^i$ and $\langle \tilde{\boldsymbol{\varepsilon}} \rangle^i, i = 1, 2, 3$, represent the obtained averaged tensors for each case. The system of equations (2.53) can be broken down into matrix form:

$$\begin{bmatrix} \mathbf{K} & & \\ & \mathbf{K} & \\ & & \mathbf{K} \end{bmatrix} \begin{bmatrix} \mathbf{c}_1 \\ \mathbf{c}_2 \\ \mathbf{c}_3 \end{bmatrix} = \begin{bmatrix} \mathbf{s}_1 \\ \mathbf{s}_2 \\ \mathbf{s}_3 \end{bmatrix}, \quad (2.54)$$

where $\mathbf{K}$, $\mathbf{c}_1$, $\mathbf{c}_2$, $\mathbf{c}_3$, $\mathbf{s}_1$, $\mathbf{s}_2$ and $\mathbf{s}_3$ are defined as follows:

$$\begin{aligned} \mathbf{K} &= \begin{bmatrix} E_{11}^1 & E_{22}^1 & 2E_{12}^1 \\ E_{11}^2 & E_{22}^2 & 2E_{12}^2 \\ E_{11}^3 & E_{22}^3 & 2E_{12}^3 \end{bmatrix}, \\ \mathbf{c}_1 &= \begin{bmatrix} C_{1111}^* \\ C_{1122}^* \\ C_{1112}^* \end{bmatrix}, \quad \mathbf{c}_2 = \begin{bmatrix} C_{2211}^* \\ C_{2222}^* \\ C_{2212}^* \end{bmatrix}, \quad \mathbf{c}_3 = \begin{bmatrix} C_{1211}^* \\ C_{1222}^* \\ C_{1212}^* \end{bmatrix}, \\ \mathbf{s}_1 &= \begin{bmatrix} \langle \sigma_{11} \rangle^1 \\ \langle \sigma_{11} \rangle^2 \\ \langle \sigma_{11} \rangle^3 \end{bmatrix}, \quad \mathbf{s}_2 = \begin{bmatrix} \langle \sigma_{22} \rangle^1 \\ \langle \sigma_{22} \rangle^2 \\ \langle \sigma_{22} \rangle^3 \end{bmatrix}, \quad \mathbf{s}_3 = \begin{bmatrix} \langle \sigma_{12} \rangle^1 \\ \langle \sigma_{12} \rangle^2 \\ \langle \sigma_{12} \rangle^3 \end{bmatrix}. \end{aligned} \quad (2.55)$$

Our goal is to express $\mathbf{c}_1$, $\mathbf{c}_2$ and $\mathbf{c}_3$ from each respective equation to obtain the elements of the effective elastic tensor $\tilde{\mathbf{C}}^*$. This system of equations can be solved analytically or numerically.

2.4.4 Effective elastic tensor for specific load matrices

By specifically choosing the matrices $\mathbf{E}_1$, $\mathbf{E}_2$, $\mathbf{E}_3$, the elements of the tensor $\tilde{\mathbf{C}}^*$ can be obtained explicitly. Next, we consider the following three linearly independent

load matrices:

$$\mathbf{E}_1 = \begin{bmatrix} \beta & 0 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{E}_2 = \begin{bmatrix} 0 & 0 \\ 0 & \beta \end{bmatrix}, \quad \mathbf{E}_3 = \begin{bmatrix} 0 & \beta \\ \beta & 0 \end{bmatrix}, \quad (2.56)$$

where $\beta \neq 0$ is the load parameter that corresponds to the relative magnitude of the load or displacement.

These load matrices can be equivalently expressed using Voigt notation (2.15):

$$\tilde{\mathbf{E}}_1 = \begin{bmatrix} \beta \\ 0 \\ 0 \end{bmatrix}, \quad \tilde{\mathbf{E}}_2 = \begin{bmatrix} 0 \\ \beta \\ 0 \end{bmatrix}, \quad \tilde{\mathbf{E}}_3 = \begin{bmatrix} 0 \\ 0 \\ 2\beta \end{bmatrix}. \quad (2.57)$$

Substituting (2.57) into the relations (2.52), (2.55), we get:

$$\begin{aligned} C_{1111}^* &= \frac{\langle \sigma_{11} \rangle^1}{\beta}, & C_{1122}^* &= \frac{\langle \sigma_{11} \rangle^2}{\beta}, & C_{1112}^* &= \frac{\langle \sigma_{11} \rangle^3}{2\beta}, \\ C_{2211}^* &= \frac{\langle \sigma_{22} \rangle^1}{\beta}, & C_{2222}^* &= \frac{\langle \sigma_{22} \rangle^2}{\beta}, & C_{2212}^* &= \frac{\langle \sigma_{22} \rangle^3}{2\beta}, \\ C_{1211}^* &= \frac{\langle \sigma_{12} \rangle^1}{\beta}, & C_{1222}^* &= \frac{\langle \sigma_{12} \rangle^2}{\beta}, & C_{1212}^* &= \frac{\langle \sigma_{12} \rangle^3}{2\beta}. \end{aligned} \quad (2.58)$$

2.4.5 Boundary conditions for specific load matrices

If we use the load matrices defined in (2.56), we obtain the following relations from (2.51) for the respective boundary conditions::

$$\begin{aligned} \mathbf{u}_{D1} = \mathbf{E}_1 \mathbf{x} &\iff \mathbf{u}_{D1} = \begin{bmatrix} \beta & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \beta x_1 \\ 0 \end{bmatrix}, \\ \mathbf{u}_{D2} = \mathbf{E}_2 \mathbf{x} &\iff \mathbf{u}_{D2} = \begin{bmatrix} 0 & 0 \\ 0 & \beta \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ \beta x_2 \end{bmatrix}, \\ \mathbf{u}_{D3} = \mathbf{E}_3 \mathbf{x} &\iff \mathbf{u}_{D3} = \begin{bmatrix} 0 & \beta \\ \beta & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \beta x_2 \\ \beta x_1 \end{bmatrix}. \end{aligned}$$

If β is positive, it is a stretch, while a negative value corresponds to a squeeze. The resulting displacement fields $\mathbf{u}_{D1}$, $\mathbf{u}_{D2}$ and $\mathbf{u}_{D3}$ physically represent the relative displacements caused by a load of size *beta* in the horizontal, vertical and diagonal (shear) directions.

3 Computational program description

The aim of this chapter is to describe the procedures used in solving the homogenization problem of mechanics, including how they are implemented in a computer program. The **main task** of the implemented computational tool is **in the following order**:

1. Fracture and computational network generation,
2. Mechanics simulation,
3. Calculation of the effective elastic tensor.

These three main parts are dealt with in the subchapters 3.3, 3.4 and 3.5. They are preceded by two introductory subchapters: 3.1, which deals with installation and general prerequisites, and 3.2, which describes the required input parameters. In the final subchapter 3.6, the use of the program in practice and the visualization of the obtained results are demonstrated on a specific example.

3.1 Introduction to the program, installation and prerequisites

In order to use the implemented program for numerical homogenization in rock mechanics, it is first necessary to install or download several software:

- BGEM - geometry creation and meshing package (BGEM, 2025)
- Flow123d - computational simulator (Flow123d, 2025)
- GitHub repository - repository with complete code (Siddall, 2025)

It will not be described here how to work with or use these programs. Any instructions can be found in their documentation. Each of the main steps listed in the introduction of the chapter 3 corresponds to one of the **main scripts**:

- `create_mesh.py` - generates a fracture/computational network (chap. 3.3),
- `create_yaml.py` - creates a model of the mechanics (chap. 3.4),
- `create_tensor.py` - calculates the effective elastic tensor (chap. 3.5).

Each of these scripts generates specific output files: `.msh` (computational mesh), `.yaml` (simulator input) and `.txt` (effective elastic tensor). All the three steps or scripts are discussed in detail in the subsections mentioned above.

3.2 Input and configuration file

The input configuration file of the program, `config_file.yaml`, is divided into three parts, each corresponding to the input parameters of one of the main scripts: `create_mesh.py` (chap. 3.2.1), `create_yaml.py` (chap. 3.2.2) and `create_tensor.py`. 3.2.1 focus exclusively on key parameters that **directly affect** the calculations. Optional and user-defined settings, such as directories, file names, or display of results, are not included.

3.2.1 Input for artificial fracture generation and computational mesh

Most of the parameters needed for generating artificial fractures have been discussed in detail in the theoretical part of this paper, including their definitional scope (third column of table 3.1). In practice, however, these parameters take values in narrower intervals based on experimental measurements, which, as described in the introduction of the chapter 2.2, is a very challenging process.

In order to make our DFM model as faithful to reality as possible, we tried to use real experimentally measured values of the input parameters. These realistic values, or intervals, are listed in the fourth column of table 3.1 and were taken mainly from (S. Follin et al., 2006, chap. 3.4), which were obtained during research carried out in the vicinity of the *Forsmark* nuclear power plant on the east coast of Sweden.

Parameter name	Marking	Theory	SKB	Units
General geometry and fracture network settings				
<code>rectangle_dimensions</code>	$\Omega = l \times h$ ¹	$\mathbb{R}^+ \times \mathbb{R}^+$	—	$m \times m$
<code>sample_range</code>	$[r_{\min}, r_{\max}]$	$\mathbb{R}^+$	$[0.038, 169]$	m
<code>k_r</code>	k_r	$(2, \infty)$	$[2.55, 3.45]$ ²	—
<code>diam_range</code>	$[r_0, r_\infty]$	$\mathbb{R}^+$	$[0.038, 169]$	m
<code>p32_r0_to_r_infty</code>	$P_{32}[r_0, r_\infty]$	$\mathbb{R}^+$	$[0.59, 2.60]$	$\frac{m^2}{m^3}$
Statistical parameters of the fracture network				
<code>fisher_trend</code>	ϕ_{tf}	$[0, 360]$	$[2.5, 354.2]$	°
<code>fisher_plunge</code>	ϕ_p	$[0, 90]$	$[0.1, 85.4]$	°
<code>fisher_kappa</code>	κ_f	$\mathbb{R}_0^+$	$[6.5, 42.9]$	—

Table 3.1: Overview of parameters for generating artificial fractures

¹In chap. 2.3.3, Ω was defined in general, but for the calculations a rectangular region given by the Cartesian product $l \times h$ is considered.

²The range of the k_r parameter depends on the chosen procedure. This wider range is obtained by connectivity analysis and is narrower when using a tectonic continuum: $[2.55, 2.79]$.

In addition to these parameters, other positive parameters related to the generated computational network are included in the first part of the configuration file. These parameters affect the fineness of the generated finite elements (elements), determine the accuracy and tolerance in numerical calculations (e.g. `fracture_mesh_step`, `tolerance`, etc.).

3.2.2 Input for creating a file for the homogenization task

The Flow123d computational simulator used requires input data in `.yaml` format, which defines the homogenization problem to be solved. Details on the structure and settings of this file are given in (Březina, Stebel, et al., 2025, Ch. 4). However, to create the file correctly, it is necessary to specify the values of the four additional parameters that enter the homogenization task in advance:

	Name of parameter	Marking
1	<code>displacement_percentage_all_boundary_conditions</code>	β
2	<code>cross_section_multiplier</code>	η
3	<code>young_modul_rock_gpa</code>	Y_m
4	<code>reduction_value_for_fractures</code>	ξ

Table 3.2: Overview of parameters and their established labeling for the homogenization task

The first parameter corresponds directly to the value of β given in the equation (2.56).

Since the initial fracture opening before the deformation of δ actually depends not only on the displacement field but also on the size of the computational domain, it is necessary to apply the scaling of δ before the calculations. This is determined by the second parameter, η , which generally expresses the ratio of the fracture opening to its length. This is mathematically expressed as $\delta_f = \eta l_f$ (corresponding to the typical *spherical* fracture shape). However, in the context of this work it is not possible to automatically determine the lengths of individual fractures, so it was decided to compromise by relating δ at least to the dimension of the region, i.e. $\delta = \eta l$. Numerically, the value $\eta = 0.01$ is typically chosen for (Wang et al., 2024, ch. 1), which means that for a computational area of size $1 \times 1 \text{ m}^2$, the fracture spacing will be scaled to centimeters. The parameter η is mathematically limited to the interval $\eta \in (0, 1]$, but values greater than $\eta_{\max} \approx 0.1$ are not physically meaningful.

The third parameter is the Young's modulus of the matrix Y_m (GPa units), and the Young's modulus for fractures Y_f is introduced by analogy. Typical values of Y_m for different rocks are given in Table 3.3.

The Young's modulus values given in the table are approximate because they depend on the specific location, geological conditions and other factors. Different rock types may exhibit different material properties. For computational purposes, the value $Y_m = 31.5 \text{ GPa}$, which corresponds to granite, was often used in this work.

Hornina	Y_m [GPa]
Žula	31.5
Andezit	41.9
Čedič	40.0
Vápenec	25.7

Table 3.3: Youngův modul pro různé horniny, zdroj (Zhang, 2016, tab. 6.12).

The fourth parameter, xi , expresses how many times the Young's modulus of the fractures is smaller than the Young's modulus of the surrounding rock matrix, i.e. $Y_f = Y_m$. The usual values are in the range $\xi \in (100, 1000)$.

3.3 The principles of generating an artificial fracture network and computational mesh

The goal of the `create_mesh.py` method is to create a computational mesh (a file in `.msh` format), which is then used as one of the inputs for the homogenization task. The general scheme for generating the artificial fractures and the computational mesh is shown in Figure 3.1.

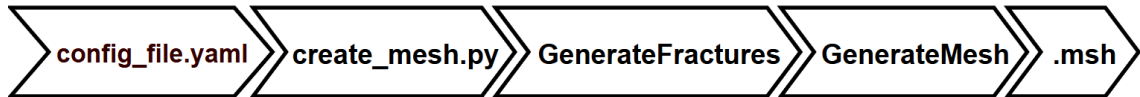


Figure 3.1: Network generation flowchart

We will not discuss the `GenerateMesh` class in detail here, it is enough to mention that it uses the implemented GMSH and BGEM functions to create and then *heal* the generated mesh. The healing process improves the quality of the generated finite element(s), leading to numerically more stable and accurate simulation results.

3.3.1 Representation of the fracture network

The resulting fracture network is represented as a set of one-dimensional segments, or individual fractures, each characterized by its location, orientation, size and membership in a particular family/population.

Each fracture belongs to a particular *family*. A family represents a set of fractures generated on the basis of the same statistical and general parameters, which are listed in table 3.1.

Families of fractures are then grouped into *populations*, which are defined not only by the individual families, but also by the general shape of the fractures (corresponding to the segments) and the computational domain. Populations typically contain multiple distinct families oriented in different directions, allowing for a more realistic simulation of the natural distribution of fracture networks in a rock mass. In this work, however, it was decided that it was sufficient to generate only one fracture family for modelling purposes.

3.3.2 Application of a fracture network for computational network generation

The process of generating artificial fractures takes place in the `GenerateFractures` class and starts with the creation of a rectangular computational domain and its boundary using GMSH. Random fractures are then added to this domain, whose geometry and orientation are based on the specified statistical parameters.

As part of this process, the three-dimensional fracture network is projected onto a planar two-dimensional domain, which is achieved by setting the z -components for the corresponding internal parameters of the fractures to zero. Next, the so-called

fragmentation is automatically performed, where the fractures and the boundaries of the computational domain are split at their intersections. These objects are then divided into several physical regions (regions), i.e. into the fractures themselves with their intersections and the boundaries of the region with the fracture intersections.

3.4 Creating a mechanics model

The input file for the Flow123d simulator is automatically generated by the `GenerateYaml` class, which is in the `create_yaml.py` script. A detailed description of setting input files and parameters for the Flow123d computational simulator is given in (Březina, Stebel, et al., 2025, ch. 4), where all formal details are given.

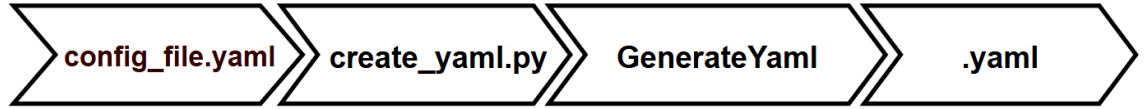


Figure 3.2: Flowchart of the ensemble creation procedure for the homogenization task

All parameter specification for the homogenization task definition is done automatically using the method `generate_yaml.py` in the `GenerateYaml` class. This method generates a `.yaml` file with the following key parameters defined: `cross_section` (ηl), `young_modulus` (Y_m, Y_f), `poisson_ratio` ($\nu = 0$), `bc_displacement` ($\mathbf{u}_{D1}, \mathbf{u}_{D2}, \mathbf{u}_{D3}$), and `mesh_file` is the absolute path to the generated computational network (`.msh`).

The Poisson's number was set to $\nu = 0$, which eliminates the stress in the third direction. The parameter `cross_section_min` corresponds to the value of $\delta_{\min}$ (see section 2.3.4) and we also used a value of zero to simplify the calculations.

Furthermore, the parameter `contact` is set to the value `true`, which activates the application of contact conditions in the model. With contact conditions enabled, the mechanical model is solved as a quadratic programming problem with linear constraints.

3.5 Output and calculation of the elements of the resulting tensor

If the procedures described in chapters 3.3 and 3.4 have been successfully completed, we can proceed to the actual simulation using the Flow123d computational simulator. This process is implemented in the `GenerateVtuFiles` class. The output of the simulations are three different `.vtu` files, each corresponding to one boundary condition, which was described in the 2.4.5 section. The final effective elastic tensor is then calculated in the `GenerateTensor` class based on the data from these files.

The method `compute_effect_elast_constants_voigt` in the class

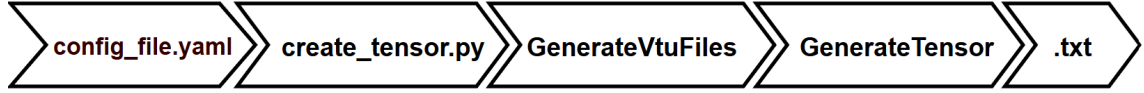


Figure 3.3: Flowchart of the procedure for creating a file with the resulting tensor

`GenerateTensor` is crucial, because it determines all the effective elastic constants, i.e. the individual elements of the tensor $\tilde{\mathbf{C}}^*$, based on the equation (2.58), which are then converted into Voigt notation.

3.6 Calculation of the sample problem

The goal of this subsection is to apply the general principles and procedures described in the previous subsections to a specific sample problem. All the data and calculations shown in this chapter are available in the `Raw_data` directory, where the `Thesis_example_problem` subdirectory is located, which is uploaded to (Siddall, 2025). We begin by listing the parameter values used to create the `.msh` and `.yaml` files that are necessary to perform the calculations and analyses in this problem:

Name of parameter	Marking	Values	Units
rectangle_dimensions	$l \times h$	1×1	m^2
sample_range	$[r_{\min}, r_{\max}]$	$[0.038, 169]$	m
k_r	k_r	2.6	—
diam_range	$[r_0, r_{\infty}]$	$[0.038, 169]$	m
p32_r0_to_r_infty	$P_{32}[r_0, r_{\infty}]$	2	$\frac{m^2}{m^3}$
fisher_trend	ϕ_{tf}	10	$^{\circ}$
fisher_plunge	ϕ_p	85	$^{\circ}$
fisher_kappa	κ_f	10	—
displacement_..._conditions	β	0.07	—
cross_section_multiplier	η	0.05	—
young_modul_rock_gpa	Y_m	31.5	GPa
reduction_value_for_fractures	ξ	500	—

Table 3.4: Parameters used for the sample task

Figure 3.4 below shows, using GMSH, what the generated artificial fractures (left) and the associated computational mesh (right) look like for a sample problem.

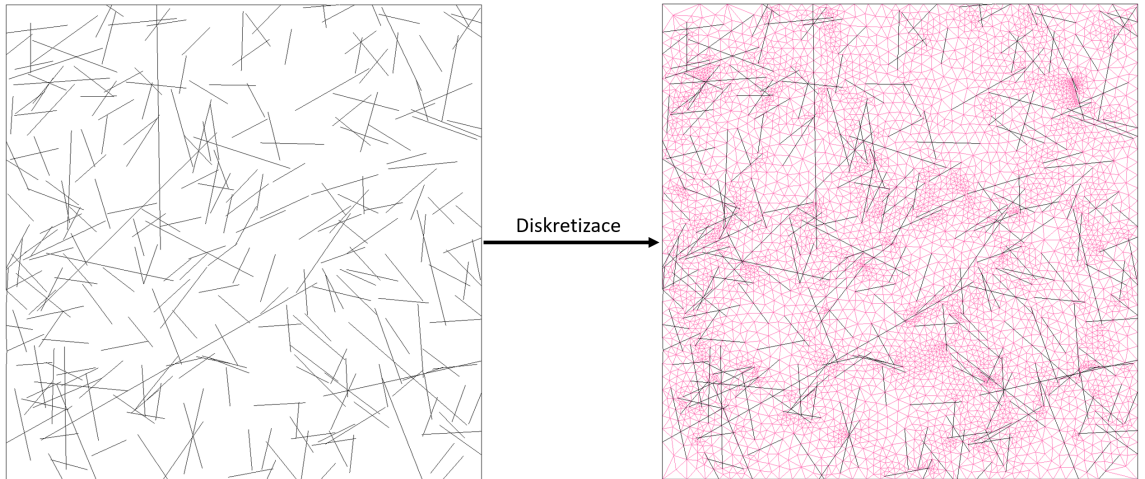


Figure 3.4: Generated artificial fractures (left) and computational mesh (right)

Next, we show the displacement vector field $\mathbf{u}$ for the first boundary condition, i.e. the displacement in the horizontal direction. This and the following images were

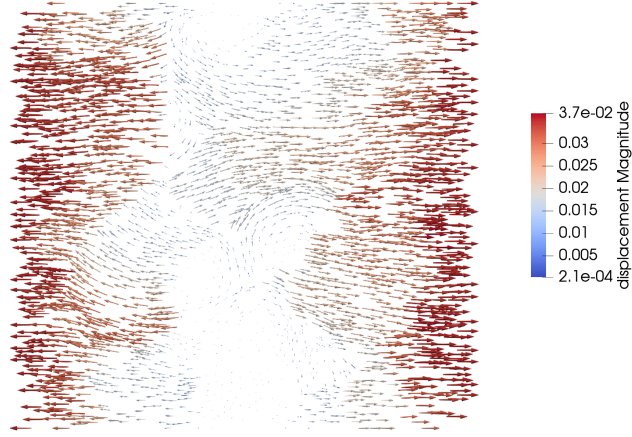


Figure 3.5: Displacement vector field for horizontal displacement

created using the visualization software *ParaView* (Ahrens et al., 2005). In the figure 3.5 the displacement vector field is shown in color, each point is described by a vector that shows the direction and magnitude of the displacement. The scale on the right (in metres) represents the magnitude of the displacement itself.

To better visualize the deformation of the fracture network itself, we can compare the model before and after loading as shown in the figure 3.6.

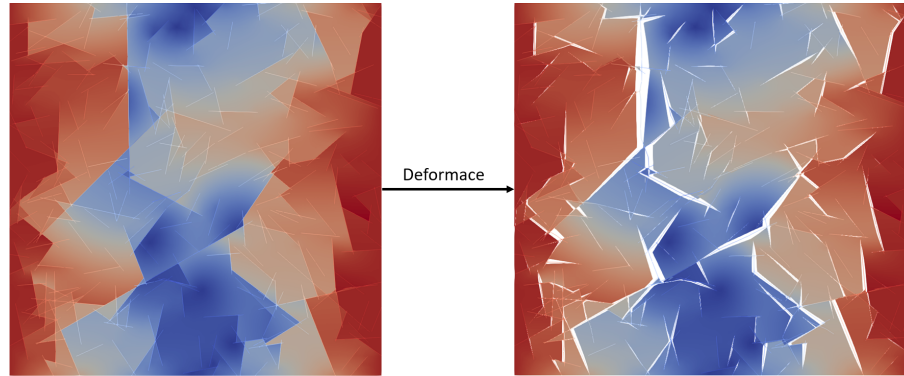


Figure 3.6: Tensile deformation in the horizontal direction in a sample problem

Finally, we show the resulting effective elastic tensor, which is shown in Figure 3.7 below.

Effective elastic tensor in matrix form for 2D problems			
C =	3205877199.349973	181230622.837998	-49703200.23166834
	182723454.08514413	2177755275.4028234	18383822.75753099
	-49762563.481876045	16804988.521057494	883477862.6823198

Figure 3.7: Resulting effective elastic tensor for the example problem

4 Stochastic homogenization and statistical analysis of results

This chapter is devoted to numerical simulations aimed at analyzing the influence of various parameters on the values of the effective elastic tensor. This has made it possible to understand the behaviour of the DFM model. Calculations have been performed to analyze the influence of these parameters:

1. orientation of fractures, κ_f ,
2. relative area of fractures, η ,
3. the stiffness of the fractures, ξ
4. displacement magnitude β ,
5. geometry of the fracture network (stochastic homogenization).

The following subchapters correspond to this order [4.1](#), [4.2](#), [4.3](#), [4.4](#) and [4.5](#), where the sub-analyses are described in more detail. In the last subsection [4.6](#) is a summary of all observations.

For all sub-analyses, constant values of parameters that were not directly analyzed were used. These parameters therefore remained the same in all analyses or simulations and their values are shown in Table [4.1](#).

Name parameter	Marking	Values	Units
rectangle_dimensions	$l \times h$	1×1	m^2
sample_range	$[r_{\min}, r_{\max}]$	$[0.038, 169]$	m
k_r	k_r	2.6	—
diam_range	$[r_0, r_{\infty}]$	$[0.038, 169]$	m
p32_r0_to_r_infty	$P_{32}[r_0, r_{\infty}]$	2	$\frac{m^2}{m^3}$
fisher_trend	ϕ_{tf}	10	$^\circ$
fisher_plunge	ϕ_p	85	$^\circ$
young_modul_rock_gpa	Y_m	31.5	GPa

Table 4.1: Values of general parameters used in all analyses

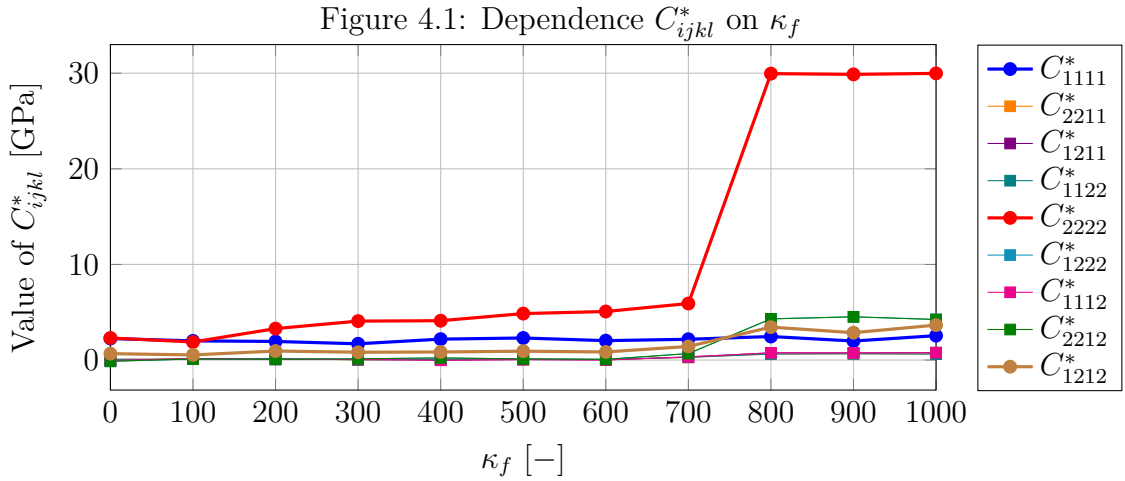
If a particular parameter was not varied, its value was constant across all simulations, ensuring consistency of results. The values used were $\kappa_f = 10$, $\eta = 0.05$, $\xi = 500$ and $\beta = 0.07$.

4.1 Dependence of the effective elastic tensor on the orientation of the fractures

The aim of this section was to analyze how the change of the parameter κ_f affects the components of the effective elastic tensor C_{ijkl}^* . Numerical simulations were performed for different values of κ_f , in the interval from 0 to 1000. For each value of κ_f , a new crack network was created.

As described in the section 2.2.2, as κ_f increases, the cracks should show a certain *dominant direction* (i.e. exactly in the direction defined by $\mathbf{a}$, which in our case is 85°), which should cause a large increase in the element C_{ijkl}^* corresponding to this direction.

The results of the calculations are shown in Figure 4.1, where the values of the individual components of the effective elastic tensor are plotted against the parameter κ_f .



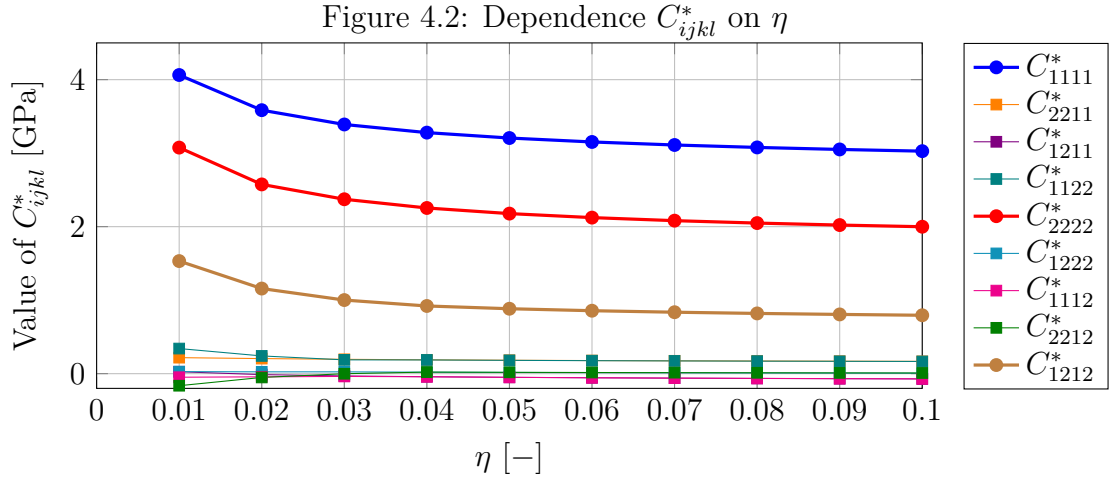
From the results of the calculations, it is clear that there was indeed a large increase in the element corresponding to the dominant fracture direction, namely C_{2222}^* , while the other elements remained relatively constant.

4.2 Dependence of the effective elastic tensor on the ratio of the fracture opening to its length

In this analysis, the aim was to investigate the effect of changing the parameter η , which represents the ratio of the crack opening to its length, on the components of the effective elastic tensor C_{ijkl}^* . Numerical simulations were performed for different values of η ranging from 0.01 to 0.1. The same computational grid was used for all calculations.

From theory, we know that as the relative area of the fractures increases, the overall stiffness of the material will gradually decrease, since the fractures always

have a lower stiffness than the surrounding rock matrix, resulting in a decrease in the values of C_{ijkl}^* . In other words, we should see decreasing values of C_{ijkl}^* as η increases.

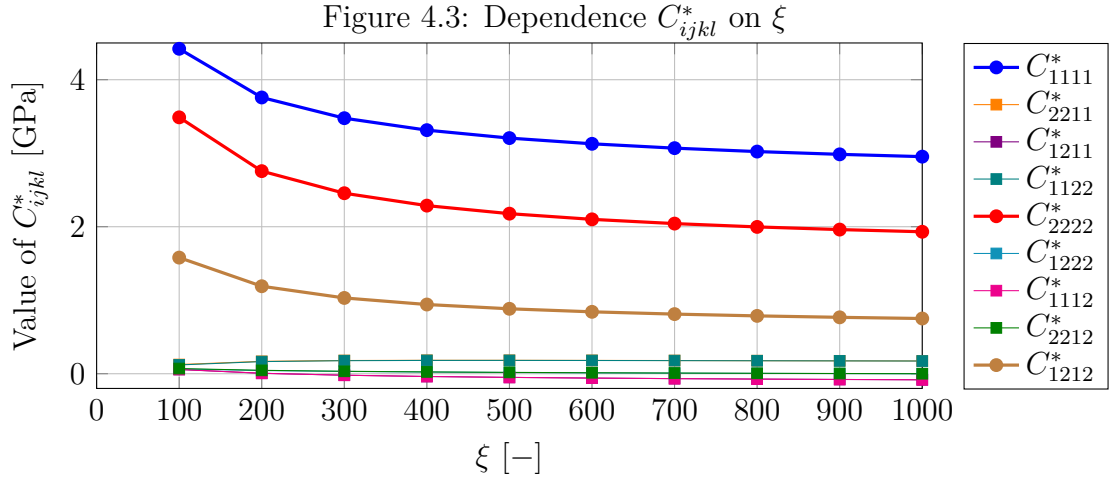


The most pronounced decrease is observed in the main diagonal components C_{1111}^* , C_{2222}^* and C_{1212}^* .

4.3 Dependence of the effective elastic tensor on fracture stiffness

Simulations were performed for different values of ξ in the interval from 100 to 1000. For all calculations, a single fracture network was used, which, among others, was also used in the previous analysis.

It is natural to assume that the value of the effective stiffness tensor C_{ijkl}^* will increase with increasing fracture stiffness. In other words, the smaller the value of the parameter ξ , the higher the overall stiffness of the material will be and hence the values of C_{ijkl}^* .

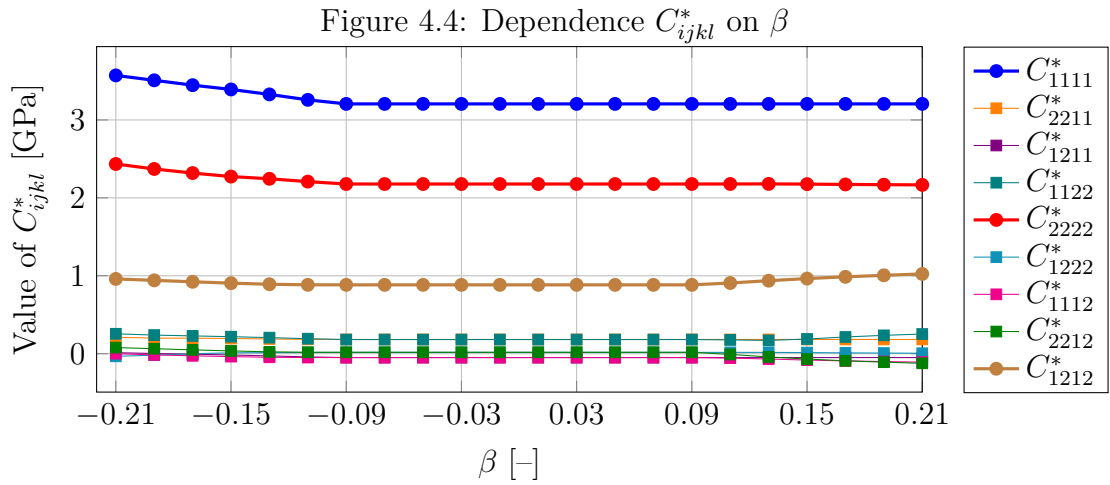


Again, we can see the main changes in the diagonal elements, while the off-diagonal ones remain mostly constant.

4.4 Dependence of the effective elastic tensor on the displacement

In this section, we analyze how the effective elastic tensor changes as a function of the magnitude of the prescribed displacement. If our model were purely linear elastic and contained only a continuous medium without cracks, the resulting effective parameters would be independent of the magnitude of the load. In other words, C_{ijkl}^* would remain constant regardless of the magnitude of the prescribed displacement.

However, our DFM model includes contact terms, see (2.45). Thus, we should observe that as the negative displacement (compression) increases, some cracks start to close gradually, which will result in a nonlinear behavior of the whole model.



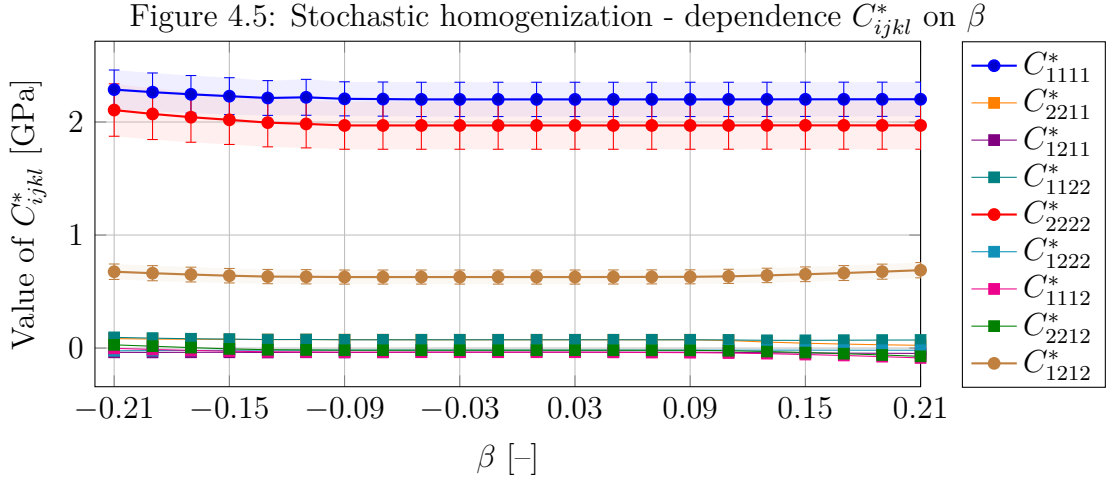
And indeed, as the results in Figure 4.4 show, the effective elastic parameters increase proportionally with increasing compression, confirming the expected nonlinear behaviour due to contact conditions.

4.5 Dependence of the effective elastic tensor on the geometry of the fracture network

This last analysis is directly related to the previous one. In principle, it is the same analysis, but performed for several different fracture networks. Since each generated network is random, the resulting values of the effective elastic tensor differ between the networks. The goal of this section is to verify that the C_{ijkl}^* values in the **average** are **independent** of a particular fracture network.

The analysis was performed for 10 different randomly generated networks with the same input (statistical) parameters. The individual components of the effective tensor were averaged (arithmetic mean) and, in addition, the standard error for the diagonal elements was determined in the form $\frac{s}{\sqrt{n}}$, where s is the sampling standard deviation and n is the sampling range. For the off-diagonal elements, the standard

errors were neglected. This process is called **stochastic homogenization** and its result is shown in Figure 4.5 below:



Since the average values for the different networks are virtually the same, we can conclude that the average effective elastic tensor as calculated in this analysis is, on average, independent of the particular implementation of the fracture network. This suggests that randomness in the geometry of the fracture network does not have a significant effect on the overall behavior of the model.

This is confirmed, among other things, by the fact that the standard errors for the diagonal elements are relatively small (less than 10%) indicating small deviations from the *actual* values of ${}^3C^*$.

As a result, we can say that the stochastic homogenization confirmed the independence (invariance) of the effective material properties or elements of C_{ijkl}^* on a particular network of fissures.

4.6 Summary of observed findings

We analyzed the effect of various parameters on the effective elastic tensor. This analysis allowed us to describe the behavior of the DFM model and showed that the results of the calculations are in agreement with theoretical predictions, which could be summarized as follows:

- **Orientation of fractures** (κ_f): A larger value of κ_f significantly increases the value of the corresponding diagonal element $\tilde{C}^*$, so the orientation of the cracks affects the values of $\tilde{C}^*$.
- **The ratio of the fracture opening to its length** (η): An increase in the ratio of the crack opening to its length results in a decrease in tensor values $\tilde{C}^*$.
- **Stiffness of fractures** (ξ): Higher crack stiffness leads to higher values of the effective elastic tensor.

- **Displacement** (β): The nonlinear characteristics of the DFM model and the effect of contact conditions were confirmed during compression.
- **Geometry**: Stochastic homogenization confirmed the average independence of $\tilde{C}^*$ values on a particular fracture network.

5 Conclusion

The aim of this bachelor thesis was to theoretically describe and model the mechanical behaviour of crystalline rocks with randomly generated fracture networks.

In the theoretical part of the thesis, the basic principles of elastic body mechanics and the mathematical description of fracture networks were described. The possibilities of statistical description of fractures and methods of their modeling were summarized. Among other things, a mathematical model combining discrete fractures and a continuous rock matrix was presented. Finally, the principles of numerical homogenization as a method to solve problems in heterogeneous materials were described.

In the practical part, a program was implemented to create two-dimensional rock models with a randomly generated fracture network according to a chosen statistical distribution and other parameters. Mechanics problems were solved on these models using numerical simulations with the inclusion of contact conditions. Various analyses, including stochastic homogenization, were performed to investigate the influence of individual parameters and the behaviour of the model as a whole.

This fulfilled all the objectives set out in the terms of reference. The results of the calculations confirmed the theoretical assumptions and showed a significant influence of fracture networks on the mechanical properties of crystalline rocks.

The motivation for this work was the understanding of the problem of heterogeneous materials with random fractures, their mathematical description and a method to practically obtain effective mechanical properties in two dimensions. A natural progression would be to extend the methods used to three dimensions, which would allow a more realistic analysis of the effect of fractures on the mechanical properties of the material.

Bibliography

- AHRENS, James, Berk GEVECI, and Charles LAW, 2005. ParaView: An End-User Tool for Large Data Visualization. In: *Visualization Handbook*. Elsevier. ISBN 9780123875822.
- BENSOUSSAN, Alain, Jacques-Louis LIONS, and George PAPANICOLAOU, 2011. *Asymptotic analysis for periodic structures*. Vol. 374. American Mathematical Soc.
- BERRE, Inga, Florian DOSTER, and Eirik KEILEGAVLEN, 2019. Flow in fractured porous media: A review of conceptual models and discretization approaches. *Transport in Porous Media*. Vol. 130, no. 1, pp. 215–236.
- BGEM, 2025. *BGEM*. Available also from: <https://github.com/GeoMop/bgem>. [Online; navštíveno 28. 03. 2025].
- BRDIČKA, Miroslav, Ladislav SAMEK, and Bruno SOPKO, 2005. *Mechanika kontinuua*. Academia.
- BŘEZINA, Jan and Jan STEBEL, 2024. Discrete fracture-matrix model of poroelasticity. *ZAMM - Journal of Applied Mathematics and Mechanics / Zeitschrift für Angewandte Mathematik und Mechanik*. Vol. 104, no. 4, e202200469. Available from DOI: <https://doi.org/10.1002/zamm.202200469>.
- BŘEZINA, Jan, Jan STEBEL, et al., 2025. *Dokumentace Flow123d*. Available also from: http://flow.nti.tul.cz/packages/3.9.0/flow123d_3.9.0_doc.pdf. [Online; navštíveno 30. 03. 2025].
- DAVIS, John C, Robert J SAMPSON, et al., 2002. *Statistics and data analysis in geology, Third edition*. Vol. 638. Wiley New York.
- FIGUEIREDO, Adelaide, 2009. Discriminant analysis for the von Mises-Fisher distribution. *Communications in Statistics-Simulation and Computation*. Vol. 38, no. 9, pp. 1991–2003.
- FLOW123D, 2025. *Flow123d*. Available also from: <https://flow123d.github.io/>. [Online; navštíveno 28. 03. 2025].
- FOLLIN, S. et al., 2006. *Hydrogeological DFN Modelling Using Structural and Hydraulic Data from KLX04: Preliminary Site Description Laxemar Subarea - Version 1.2*. SKB. SKB rapport. Available also from: <https://books.google.cz/books?id=xbCFNwAACAAJ>.

- HILL, Rodney, 1963. Elastic properties of reinforced solids: some theoretical principles. *Journal of the Mechanics and Physics of Solids*. Vol. 11, no. 5, pp. 357–372.
- HUR, Vera Mikyoung, 2009. *The Dirac Distribution*. Available also from: https://ocw.mit.edu/courses/18-034-honors-differential-equations-spring-2009/resources/mit18_034s09_lec23/. [Online; navštíveno 02. 01. 2025].
- KOUZNETSOVA, Varvara, WAM BREKELMANS, and FPT1005 BAAIJENS, 2001. An approach to micro-macro modeling of heterogeneous materials. *Computational mechanics*. Vol. 27, no. 1, pp. 37–48.
- MAPTEK PTY LTD, 2024. *Glossary*. Available also from: <https://help.maptek.com/pointstudio/2022.1/glossary.htm>. [Online; navštíveno 15.12.2024].
- MOULINEC, Hervé and Pierre SUQUET, 1998. A numerical method for computing the overall response of nonlinear composites with complex microstructure. *Computer methods in applied mechanics and engineering*. Vol. 157, no. 1-2, pp. 69–94.
- MUNIER, Raymond, 2007. *Demonstrating the efficiency of the EFPC criterion by means of Sensitivity analyses*. Tech. rep. Swedish Nuclear Fuel and Waste Management Co.
- NEČAS, Jindřich, 2011. *Direct methods in the theory of elliptic equations*. Springer Science & Business Media.
- OEHMAN, Johan and Sven FOLLIN, 2010. *Site investigation SFR. Hydrogeological modelling of SFR. Model version 0.2*.
- POVEY, Rhys G, 2023. Voigt transforms.
- SIDDALL, Ronald Ch., 2025. *Odkaz na GitHub repozitář s kódem*. Available also from: <https://github.com/RonaldSiddall/Homogenization-Rock-Mechanics-Thesis>. [Online; navštíveno 28. 03. 2025].
- STEBEL, Jan et al., 2024. On the parallel solution of hydro-mechanical problems with fracture networks and contact conditions. *Computers & Structures*. Vol. 298, p. 107339.
- WALDRON, John and Morgan SNYDER, 2020. *Geological Structures: A practical introduction*.
- WANG, Xiaolin et al., 2024. Empirical prediction of hydraulic aperture of 2D rough fractures: a systematic numerical study. *Frontiers of Earth Science*. Vol. 18, no. 3, pp. 579–597.
- ZHANG, Lianyang, 2016. *Engineering properties of rocks*. Butterworth-Heinemann.
- ZOHDI, T. I. and P. WRIGGERS, 2005. *An Introduction to Computational Micromechanics*. Springer.